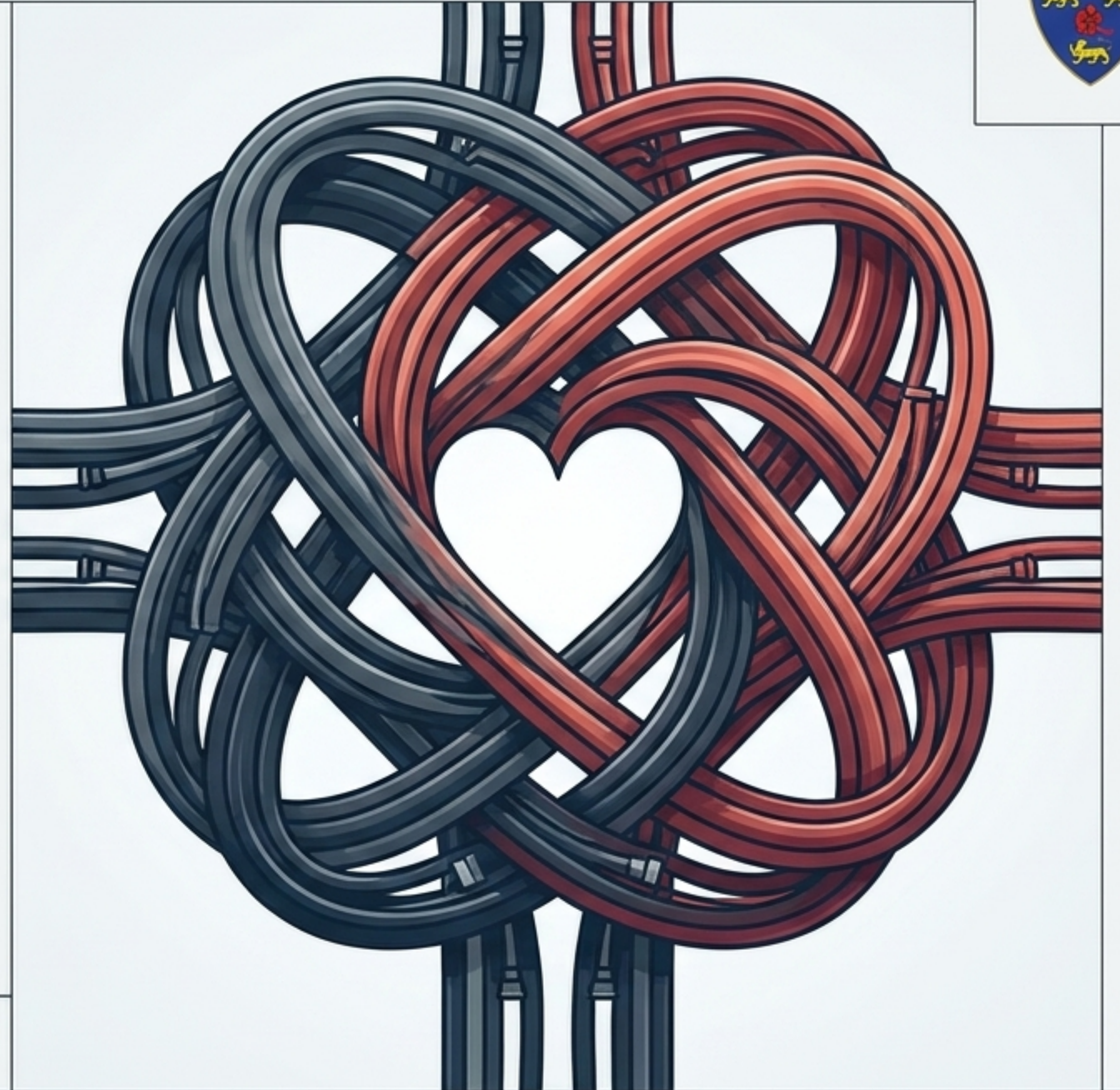




Tying the Data Knot

A State-of-the-Art
Machine Learning Pipeline
for Digital Romance

V8 Patched Edition



Engineering a Production-Grade Pipeline

Modular Architecture & Parallel Execution

Chew Wei Jian
Project Lead & ML
Pipeline Manager
Focus: Parallel Compute



Ng Jin Ru
EDA Analyst
Focus: Target Balance &
Diagnostics

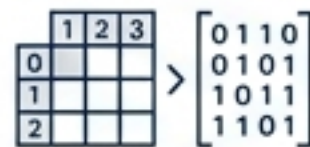


Chaang Wai Chiu
Explainability & Ethics
Focus: SHAP &
Demographic Parity

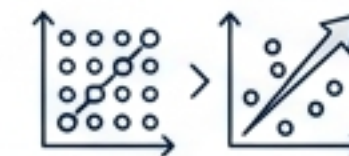


Master Pipeline

Ku Jian Cheng
Data Preprocessing &
Feature Engineer
Focus: Ordinal/Multi-Hot



Ang Ying En
Model Optimization &
Tuning
Focus: RandomizedSearchCV

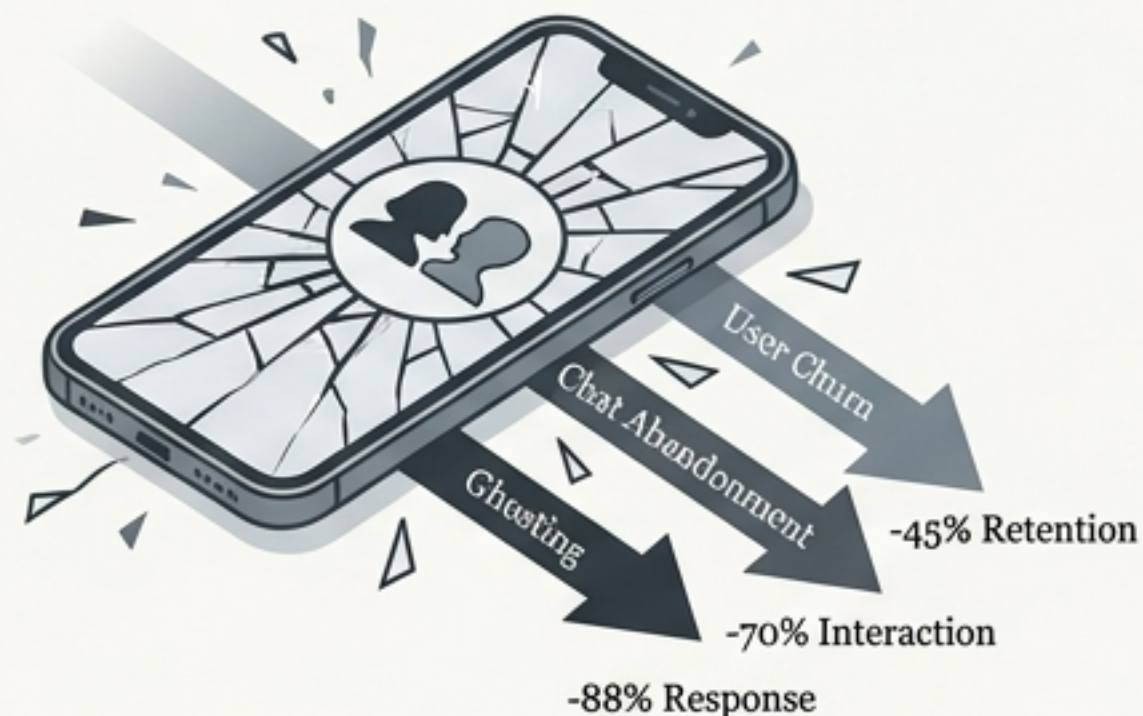


Workflow: GitHub Centralized Repo → Modular Jupyter Branches → Peer-Reviewed Merges

Foundation for V8 Architecture

The Confounding Loop of Modern Matchmaking

The Flaw



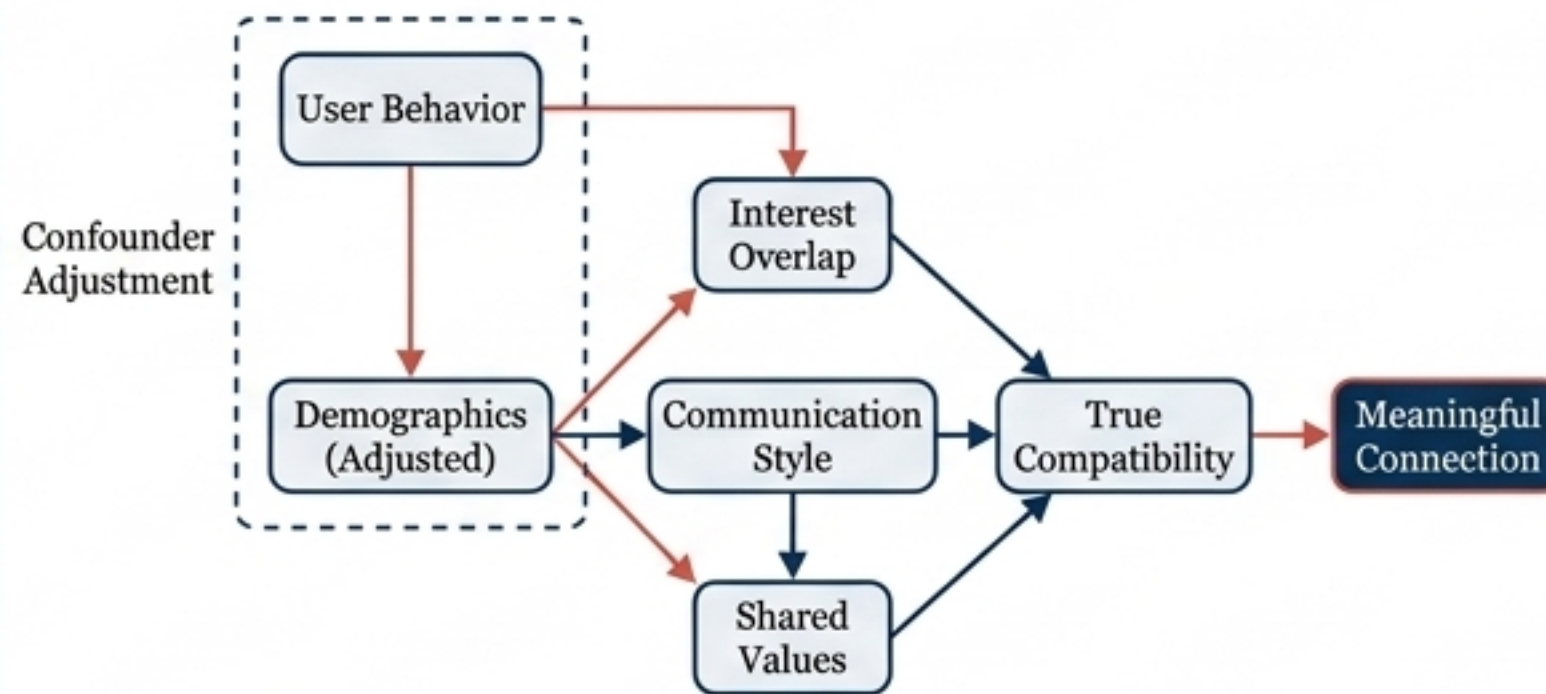
Superficial Matching
Algorithms confuse correlation with compatibility.

The Result
Skyrocketing user churn, chat abandonment, and ghosting.

The Root Cause
Latent confounders (e.g., location, income) creating backdoor bias.

*Transitioning from
Predictive to Prescriptive
Causal AI*

The V8 Objective



Deconstruct the Signal
Build a leakage-free causal audit sandbox.

Isolate True Drivers
Strip away demographic bias using causal backdoor adjustments.

Predict Meaningful Connections
Prescribe true human compatibility, not just superficial matches.

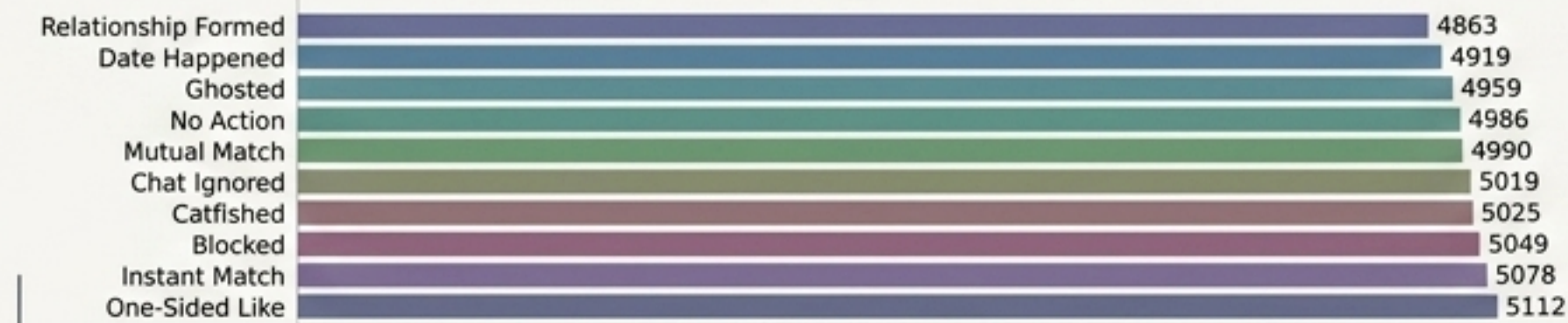
Distilling the Metric of Love

Consolidating a Sparse Multi-Class Reality into a Robust Binary Target

The Intake: 50,000 Behavioral Profiles | 25 Raw Features | 0 Missing Values

The Noise (Multi-Class Reality)

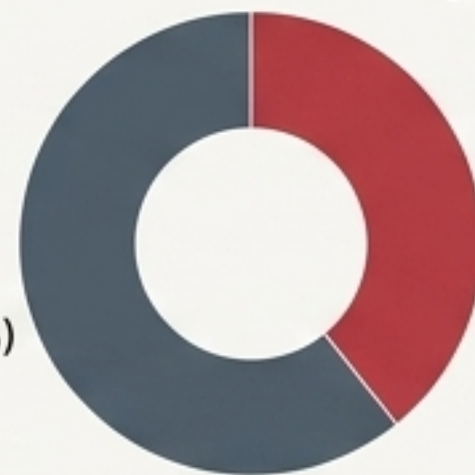
Distribution of All 10 Match Outcomes



The Distillation (Binary Target)

Negative Target (0)
Ghosted, Blocked,
Catfished, Chat Ignored,
No Action, One-Sided
Like

Negative Target (0) (60.3%)
30,150



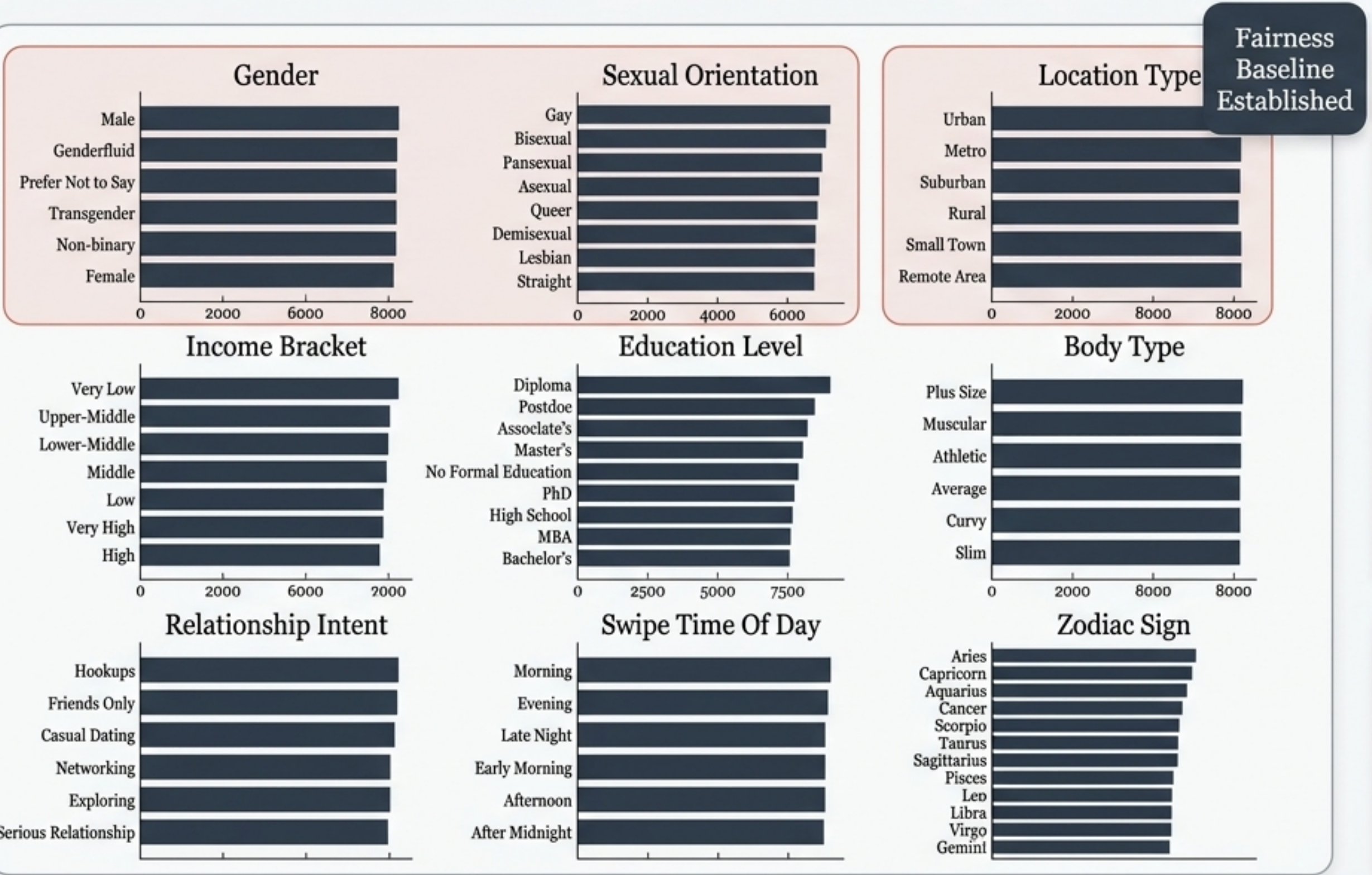
Positive Target (1) (39.7%)
19,850

Positive Target (1)
Mutual Match,
Instant Match, Date
Happened, Relationship
Formed

Result: A realistic 60/40 class imbalance typical of online matchmaking ecosystems

The Fairness Baseline: Categorical Uniformity

Programmatic distributions prevent demographic dominance in the loss function.



Key Insight: Perfect Balance
Uniform representation across Gender (6 identities), Sexual Orientation (8 types), and Location.

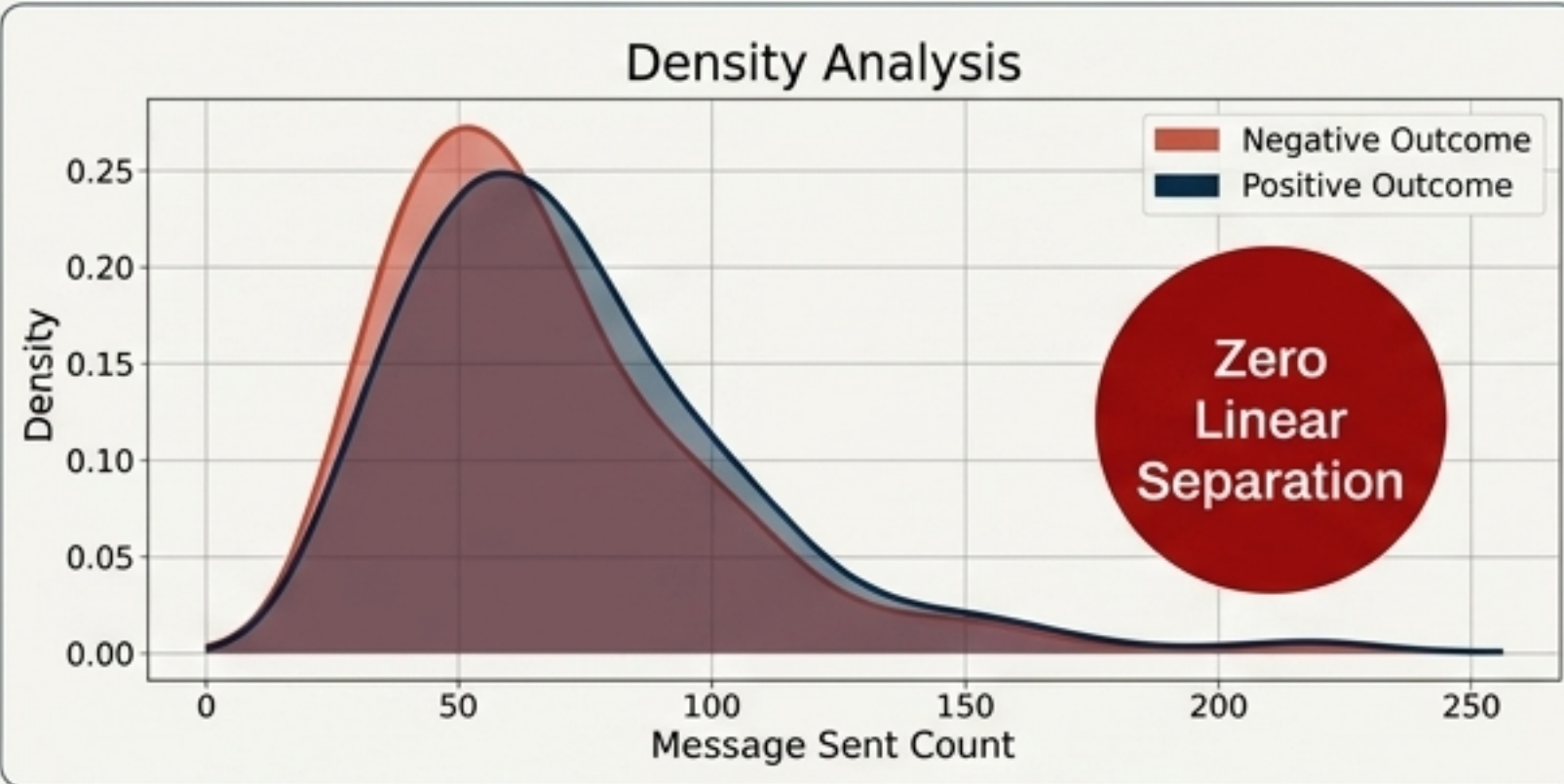
Mathematical Impact
Prevents majority-class dominance during gradient updates. Ensures estimators evaluate all demographic cohorts with equal weight.

Ethical Advantage
Provides a mathematically clean slate for downstream **Demographic Demographic Parity** and **True Positive Rate (TPR)** fairness audits.

Signal Obscurity & Power Users

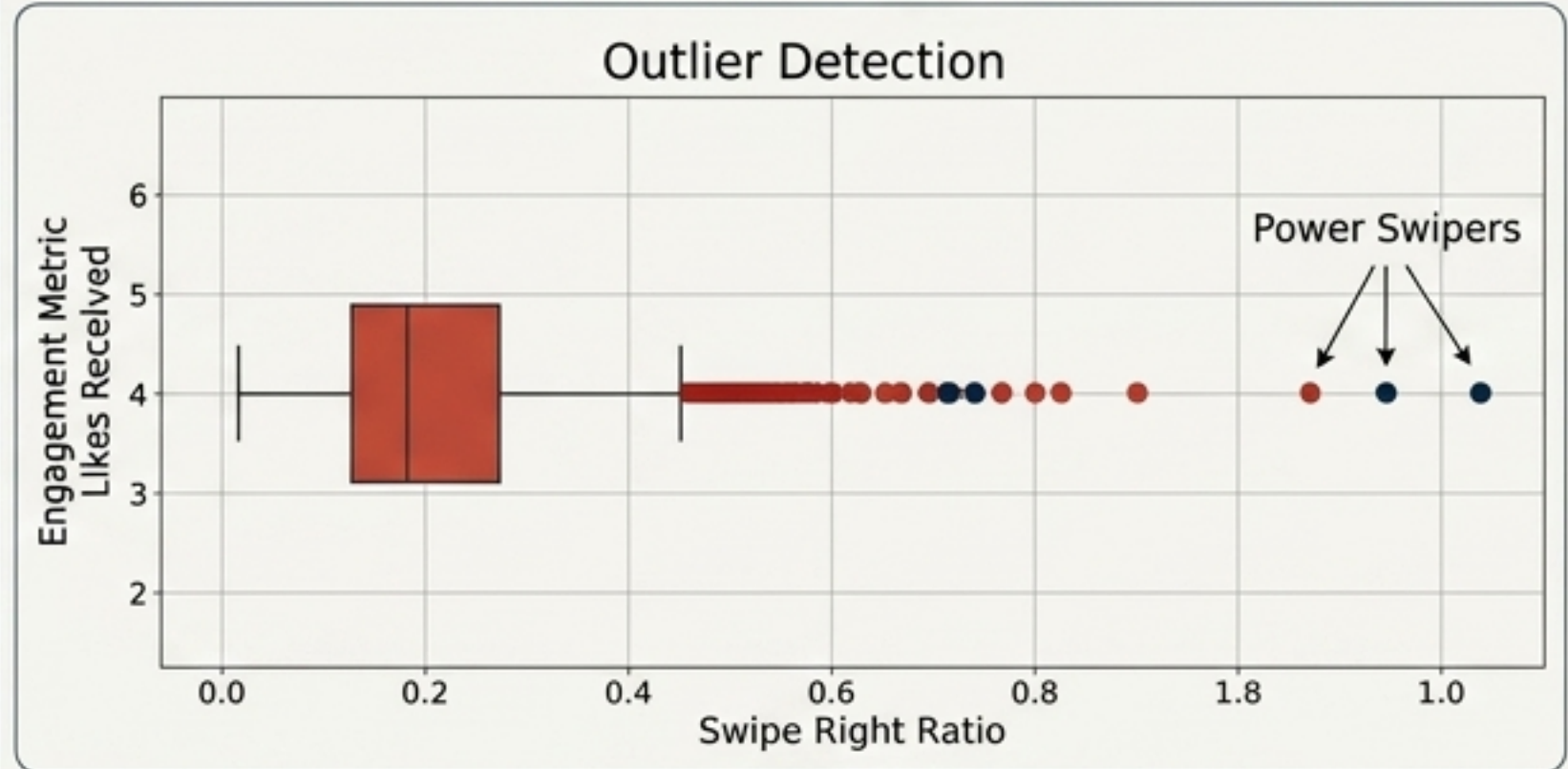
Why naive linear models and standard scaling will mathematically fail.

Signal Obscurity



Density curves for positive and negative outcomes overlap almost completely. Insight: No single continuous variable can linearly separate a successful match. Deep, non-linear multi-feature interactions are required.

Power Swipers



Boxplots reveal extreme “Power Swipers” exceeding the 1.5 IQR threshold. Action: Standard scaling would collapse under this variance. We mandate an upgrade to RobustScaler (median & IQR based).

PIPELINE UPGRADE: RobustScaler + Unsupervised Isolation Forest (OOD Filter)